



Study on the durability of GFRP bars and carbon/glass hybrid fiber reinforced polymer (HFRP) bars aged in alkaline solution

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ABSTRACT

Using glass fiber reinforced polymer (GFRP) bars as the reinforcement of concrete structures is becoming more and more popular. Carbon/glass hybrid-fiber reinforced polymer bars (HFRP bars), formed by wrapping a thin layer of carbon fiber reinforced polymer (CFRP) on GFRP bars, is expected to have better durability than GFRP bars. In order to evaluate the effects of CFRP layer on the durability of GFRP bars, the water absorption behaviors of both GFRP and HFRP bars were characterized, and the interlaminar shear strength degradation of GFRP and HFRP bars aged in alkaline solution was investigated with short beam shear test. The results indicate that after being aged in alkaline solution, the remaining interlaminar shear strengths of HFRP bars were larger than that of GFRP bars. And, the deterioration of HFRP bars aged in alkaline solution was found occurring at the CFRP/GFRP interface. The long-term interlaminar shear strength of GFRP and HFRP bars aged in alkaline solution was predicted using the Arrhenius theory. The present study shows that the CFRP layer enhances the durability of GFRP bars in alkaline environment, due to the delayed water and OH⁻ diffusion in CFRP and the better alkaline resistance of CFRP.

1. Introduction

Glass fiber reinforced polymer (GFRP) bars have been successfully used as reinforcement in highway bridges and marine structures [1]. And, adopting GFRP bars as the primary reinforcement of concrete structures is becoming more and more popular [2]. However, GFRP bars embedded in concrete, would show performance degradation as a result of the concrete pore solution which is alkaline in nature [3–5].

Lots of previous publications have shown that the mechanical properties of GFRP bars immersed in alkaline solution degrade gradually. And, the degradation level depends on the immersion time. Chen et al [6] reported that GFRP bars showed 2% and 10% reduction in tensile strength respectively after being immersed in alkaline solution (pH = 12.7 at 20 °C) for 30 days and 120 days. Benmokrane et al [7] reported that, after being immersed in alkaline solution at 60 °C for 42 days and 208 days, the apparent horizontal shear strength of GFRP was decreased by 1% and 11% respectively. The degradation of tensile strength of GFRP bars increases with exposure temperature. The alkaline solution with pH = 12.7 at 20 °C reduces the tensile strength of GFRP bars more 32% than that at 20 °C after 120 days [6]. The

mechanical properties degradation of GFRP in alkaline solution also depends on the concentration (pH) of alkaline solution. It was reported that GFRP bars showed 57% and 9% reduction in tensile strength respectively after immersion in pH 13.6 and pH 12.7 alkaline solution for 120 days at 20 °C [6], and showed 11% and 7% reduction in apparent horizontal shear strength respectively after immersion in pH 13.4 and pH 12.7 alkaline solution for 63 days at 25 °C [8].

The degradation of GFRP bars in alkaline environment mainly results from the deterioration of glass fiber and the interface between glass fiber and polymer matrix. Compared to the significant deterioration of glass fiber and the fiber–matrix interface in alkaline environment, FTIR results showed that there is almost no change in the amount of hydroxyl groups in vinyl-ester and epoxy resin matrix [7,9]. Deterioration of glass fibers is mainly caused by the leaching due to water and the etching due to free OH⁻ [10]. Free OH⁻ in alkaline solution aggressively breaks the Si–O–Si bond of glass fibers, while hydroxylation leads to pitting and roughness in fiber surface. The fiber–matrix interface in GFRP is a nonhomogeneous region of around 1 μm thick. And, the deterioration of interface is always more severe than glass fiber [10]. With presence of water and OH⁻ at the interface,

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cracks, debonding and delamination would be formed at the interface, leading to deterioration of the interfacial bonding between fiber and matrix [11].

The degradation of GFRP highly depends on the water uptake and OH⁻ diffusion process [12]. Two ways to improve the durability of GFRP in alkaline environment is to slow the water uptake and OH⁻ diffusion [13,14] and to employ fiber with better alkali resistance [15,16]. As carbon fiber shows better alkali resistance when comparing with glass fiber [17], wrapping a layer of carbon fiber reinforced polymer (CFRP) on GFRP bars to form carbon/glass hybrid fiber reinforced polymer (HFRP) is a promising way to improve the durability of GFRP bars.

The moisture absorption of GFRP, CFRP and carbon/glass HFRP bars all followed a typical Fickian curve [15]. However, the saturated water absorption of HFRP is smaller than that of GFRP but larger than that of CFRP [18]. Due to the presence of the radial GF/CF interface, HFRP showed different OH⁻ diffusion behavior compared with CFRP and GFRP [15]. Chenggao Li et al. (2019), E. Barjasteh and S.R. Nutt (2012) and Y. I. Tsai et al. (2009) studied the durability of carbon/glass hybrid-fibers reinforced polymer (HFRP) rod in various solutions [15,16,18]. Compared to GFRP and CFRP, the absorption curves of HFRP shows less differences in the diffusion rate before saturation [15]. The water uptake of CFRP (1.2%) was 20% larger than that of HFRP (1.0%), while GFRP (0.75%) absorbed 33% less than that of HFRP [18]. Compared to the radial water absorption behavior, the water molecules in the longitudinal direction of HFRP diffused faster than that of radial direction [16]. It is worth to note that most of the previous focused on the the HFRP bar formed by wrapping a layer of GFRP around CFRP.

Although numerous studies have investigated the effects of alkaline solution on the mechanical properties of GFRP and CFRP bars, few publications on the durability of carbon/glass HFRP bars are available [19–22]. There are needs to make clear the adverse effects of water molecules and hydroxyl ions on the mechanical properties of HFRP prior to applying HFRP bars in concrete structures.

Compared with GFRP, the carbon/glass HFRP showed enhanced mechanical properties, depending on the position of carbon fiber layer, as a result of the fiber hybridization effects [23]. In the present study which mainly focuses on the hybridization effects of CFRP on improving the durability of GFRP bars, carbon/glass HFRP bars were prepared by wrapping a thin layer of (approximately 0.7 mm thick) CFRP around GFRP bars, and the volume fraction of carbon fiber in HFRP is less than 18%. This is to eliminate the hybridization effects of carbon fiber on enhancing the mechanical properties of GFRP bars. In order to investigate the effects of fiber hybridization on durability of FRP, HFRP and GFRP bars were conditioned in alkaline solution, and then characterized and compared in terms of the short beam shear strength, water absorption and microstructure. Existing models were

also employed to predict the long-term performance of GFRP and HFRP bars in alkaline solution.

2. Experimental

2.1. Raw materials

Unidirectional GFRP and HFRP bars with nominal diameter of 8 mm, as shown in Fig. 1a, were used in the present study. Both FRP bars were supplied by the Harbin FRP Institute of China. Same epoxy resin, composed of Bisphenol-A resin and Methylhexahydrophthalic Anhydride hardener supplied by the Bluestar New Chemical Materials Co. Ltd (Wuxi, China) was employed as the matrix materials of GFRP and HFRP bars. The E-glass fibers used were supplied by Chongqing Polymer International Corp. (Chongqing, China), and the carbon fibers were supplied by the Toray Industries Inc. (Japan). Table 1 shows the characteristics of glass and carbon fibers. Fig. 1b shows the fabrication process of HFRP bars. During the pultrusion process for fabricating GFRP and HFRP bars, the molding temperature was kept at 180 °C and the fiber pull speed was 40 cm/min. Table 2 shows the detailed parameters of GFRP and HFRP bars. The glass transition temperature (T_g) was determined by DMA thermal analysis (Three point bending mode). The peak of tan delta was set as T_g .

2.2. Conditioning

With refer to Wang et al [20], alkaline solution of pH 13.4, which was similar to OPC based concrete pore solution, was prepared with distilled water for conditioning the GFRP and HFRP bars. The composition of the alkaline solution was shown in Table 3. The pH was tested by the pH indicator during the exposure time, and reduced to be 13.0. In the present study, GFRP bars and HFRP bars were immersed in alkaline solution at room temperature, 40 °C and 60 °C. Indoor average temperature was measured to be approximate 21 °C. And, for each case, 4 immersion periods, including 0, 30, 70, 100 and 140 days, were adopted. Fig. 2 shows the FRP bars conditioned in alkaline solution. After being exposed to alkaline solution for specified periods, HFRP and GFRP bar specimens were taken out and subject to characterization, including short beam shear test, water absorption behavior study and microstructure observation with SEM (Scanning electron microscopy) and OM (Optical microscope).

The label of FRP specimens was defined as below:

“GD0” represents the control samples (without conditioning in alkaline solution) of GFRP bars, while “HD0” represents the control samples (without conditioning in alkaline solution) of HFRP bars. For aged samples, “HT21D30” represents HFRP bars immersed in alkaline solution at 21 °C for 30 days, in which “H” means HFRP bars, “D” means conditioning days, “T” means the temperature; “GT21D30” rep-

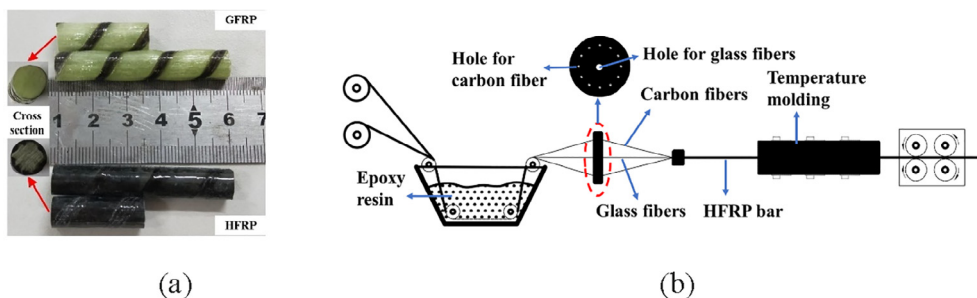


Fig. 1. GFRP and HFRP specimens (a) and forming of carbon fiber coat (b).

Table 1
Characteristics of the E-glass and carbon fibers.

Fiber	Brand/Model	Diameter (μm)	Tensile strength (MPa)	Tensile modulus (GPa)	Density (g/cm^3)
E-glass fiber	E-glass 9600 Tex	28	2200	76	2.54
Carbon fiber	Toray T700-12 K	7	4900	230	1.8

Table 2
Production parameters of GFRP and HFRP bars.

Type	V_f	T_g ($^{\circ}\text{C}$)
GFRP	62.8%	165
HFRP	–	159

Table 3
Composition and pH of the prepared alkaline solution.

Quantities (g/L)			pH
Ca(OH) ₂	NaOH	KOH	
2.0	2.4	19.6	13.4

resents GFRP bars immersed in alkaline solution at 21 $^{\circ}\text{C}$ for 30 days, in which “G” means GFRP bars.

2.3. Short beam shear test

The short beam shear test was carried out according to ASTM D4475 [24]. Fig. 3 shows the schematic sketch and image of the short beam shear test setup. The transverse diffusivity in the unidirectional FRP bars is larger than that along the fiber direction [25]. The end of FRP bars were sealed by epoxy adhesive to prevent the water diffusion along the fiber direction. Fig. 2b shows the FRP bars for the short beam shear test. The shear strength of FRPs is related to the support span-to-diameter ratio. ASTM D4475 recommends the support span-to-specimen diameter ratio from 3 to 1 up to 6 to 1. It was reported that, when the span-to-diameter ratio is in a range of 4 to 6, the effects of span to diameter on the shear strength of FRPs were insignificant [8]. The span-to-diameter ratio was hence set to be 4 in the present study. The test was conducted with an electronic universal testing

machine under displacement control mode at a displacement rate of 1.3 mm/min. The testing machine was equipped with a 25 kN loading cell. At least 4 specimens were tested for each conditioning case.

According to ASTM D4475, the interlaminar shear strength of FRPs can be determined by Eq. (1):

$$\tau_{\max} = 0.849P/D^2 \quad (1)$$

where $\tau_{\max}$ is the interlaminar shear strength of FRPs (MPa). D is the diameter of FRP bar specimen (mm). P is the peak load (kN). Fig. 4 shows the typical shear stress -deformation curves obtained in the short beam shear test. $\tau_{\max}$ is the maximum shear stress, which is defined as the apparent horizontal shear strength. s_0 is the deformation corresponding to maximum shear stress. E_0 is the initial stiffness which is defined as the load need to induce unit deformation.

2.4. Water absorption

The water absorption behavior of GFRP and HFRP bars at 21 $^{\circ}\text{C}$, 40 $^{\circ}\text{C}$ and 60 $^{\circ}\text{C}$ were studied. And, for each case, 10 specimens, with each of 50 mm long, were used. In order to determine the water absorption, the weights of specimens were measured at regular interval. For weight measurement, each specimen was taken out from water and then surface dried with tissue paper, and then the mass of specimen was measured using an electronic balance with an accuracy of ± 0.01 mg. According to ASTM D5229/D5229M [26], the water absorption of FRPs is expressed as the ratio of the mass of water absorbed by FRP specimen to the mass of the oven-dry FRP specimen, and could be determined following the Eq. (2)

$$M_i = \frac{W_i - W_0}{W_0} \times 100 \quad (2)$$

where M_i is the water absorption, W_i is the current specimen mass (g), and i represents the tested number; W_0 is the oven-dry specimen mass (g).

2.5. Scanning electron microscope (SEM) and optical microscope (OM)

The failure morphologies of FRP bar specimens subject to short beam shear test were characterized with a scanning electron micro-

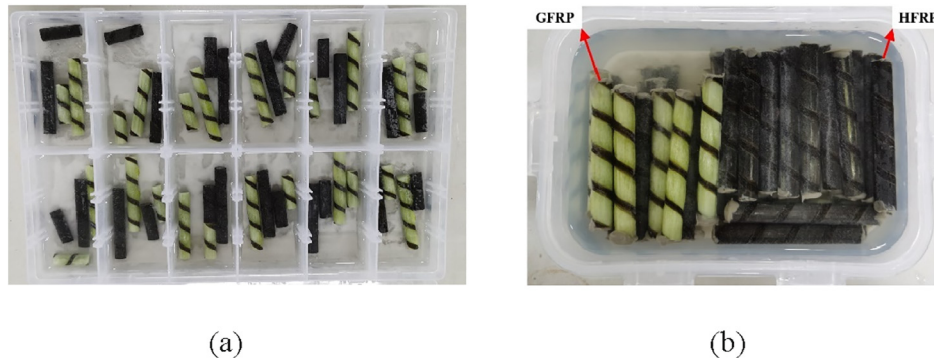


Fig. 2. FRP bars conditioned in alkaline solution for (a) water absorption and (b) shear test.

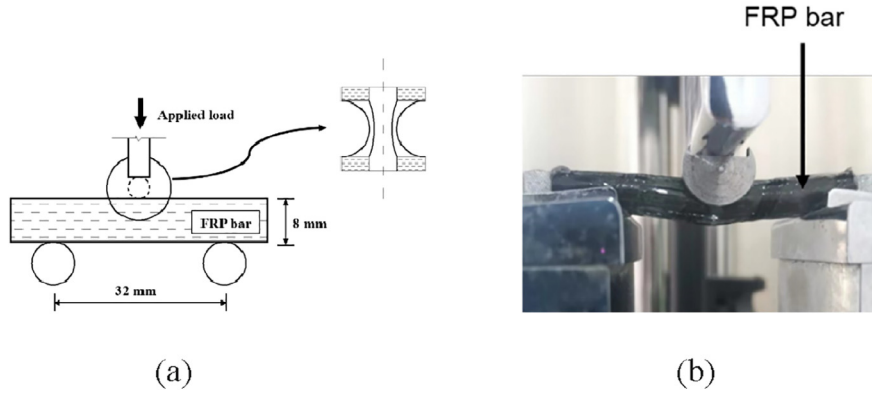


Fig. 3. Short beam shear test setup: (a) schematic sketch and (b) photograph.

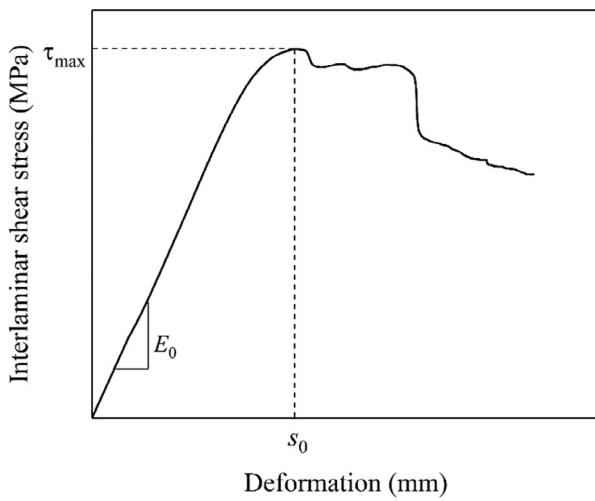


Fig. 4. Typical shear stress-deformation curve of short beam shear test.

scopy (SEM), while the cross-section of FRP bars were observed with an optical microscope (OM).

3. Results and discussion

3.1. Water absorption

Fig. 5 shows the water uptake VS immersion time curves of GFRP and HFRP bars. The water uptake of both types of FRP bars increases with increasing immersion time and is roughly proportional to the square root of immersion time. The water uptake curves of both FRP bars obtained at 60 °C achieve plateau at an immersion time of around 1776 h, which is not found in the curves obtained at 21 °C and 40 °C. This indicates that the GFRP and HFRP bars reaches saturation at 60 °C at an immersion time of around 1776 h, while both rebars do not reach saturation during the whole immersion period at 21 °C and 40 °C. The saturated water absorption in FRPs depends on the chemical structure of epoxy resin and the fiber volume fraction of FRPs [27]. The saturated water absorption of GT21, GT40 are hence similar to that of GT60, while the saturated water absorption of HT21, HT40 are similar to that of HT60.

It was reported that the weight gain of the unidirectional carbon fiber/glass fiber reinforced composite bars were coincident with the theoretical Fickian curves [15]. The water uptake behaviors of GFRP and HFRP bars follow Fickian law [28], which can be expressed as the Eq. (3) [15]:

$$M_t = \left[1 - \sum_{n=1}^{\infty} \frac{4}{r^2 \alpha_n^2} \exp(-D_c \alpha_n^2 t) \right] M_{\infty} \quad (3)$$

where M_t is the water absorption (g). M_{∞} is the saturation level of water absorption (g). t is the immersion time (s). r is the radius (mm) of FRP bar. D_c is the diffusivity coefficient (mm^2/s). α_n is the n th root of the zero order Bessel function. The Bessel function is included in the Eq. (3) is resulting from the cylindrical geometry of FRP bars. The fifty terms ($n = 50$) of summation in Eq. (3) were taken into consideration in the present study.

Table 4 shows the calculated diffusivity coefficient and the saturated water absorption of GFRP and HFRP bars. The diffusivity coefficient of both GFRP and HFRP bars increases with increasing conditioning temperature. The diffusivity coefficients of GFRP bars at 40 °C and 60 °C are 14.5 and 47.1 times larger than that at 21 °C, while the diffusivity coefficients of HFRP bars at 40 °C and 60 °C are 6.3 and 84.5 times of that at 21 °C. The activation energy of diffusion is controlled by relaxation of epoxy resin. The water molecules diffusing in epoxy resin at a larger rate at elevated temperature is because temperature rise accelerates the structure relaxation of epoxy resin [29].

Regardless of the conditioning temperature, the diffusivity coefficient of HFRP bars is smaller than that of GFRP bars. At 21 °C, 40 °C and 60 °C, CFRP coat reduces diffusivity coefficient of GFRP bars by 50%, 80% and 12%, respectively. This is because the diffusion paths of water molecules in CFRP layer is more tortuous and longer than those in GFRP, as a result of the smaller diameter of carbon fiber. According to Table 1, the diameter of carbon fiber is one-fourth of the diameter of glass fiber. As shown in Table 4, presence of a carbon fiber increases the saturated water absorption of GFRP by 19%. It can be attributed to that there are more interfacial regions in CFRP due to the smaller diameter of carbon fiber and the interfacial region between CFRP and GFRP [15].

The diffusivity coefficient (D_c) can be expressed as a function of temperature by the Arrhenius equation [30]:

$$D_c = D_0 \exp\left(-\frac{E_{ad}}{R_1 T}\right) \quad (4)$$

where E_{ad} is the activation energy of water molecular diffusion in FRPs (J/mol), R_1 is the universal gas constant (equal to 8.314 J · mol⁻¹ · K⁻¹), and D_0 is the diffusivity coefficient constant. The logarithm of D_c can be expressed as a linear function of $1/T$ with a slope of E_{ad}/R_1 ,

$$\ln(D_c) = \ln(D) - \frac{E_{ad}}{R_1 T} \quad (5)$$

Fig. 6 plots the diffusivity coefficient of GFRP and HFRP bars versus temperature, and fitting lines were made. It can be seen that, for both FRP bars, the $\ln D_c$ decreases with an increasing of $1000/T$. According

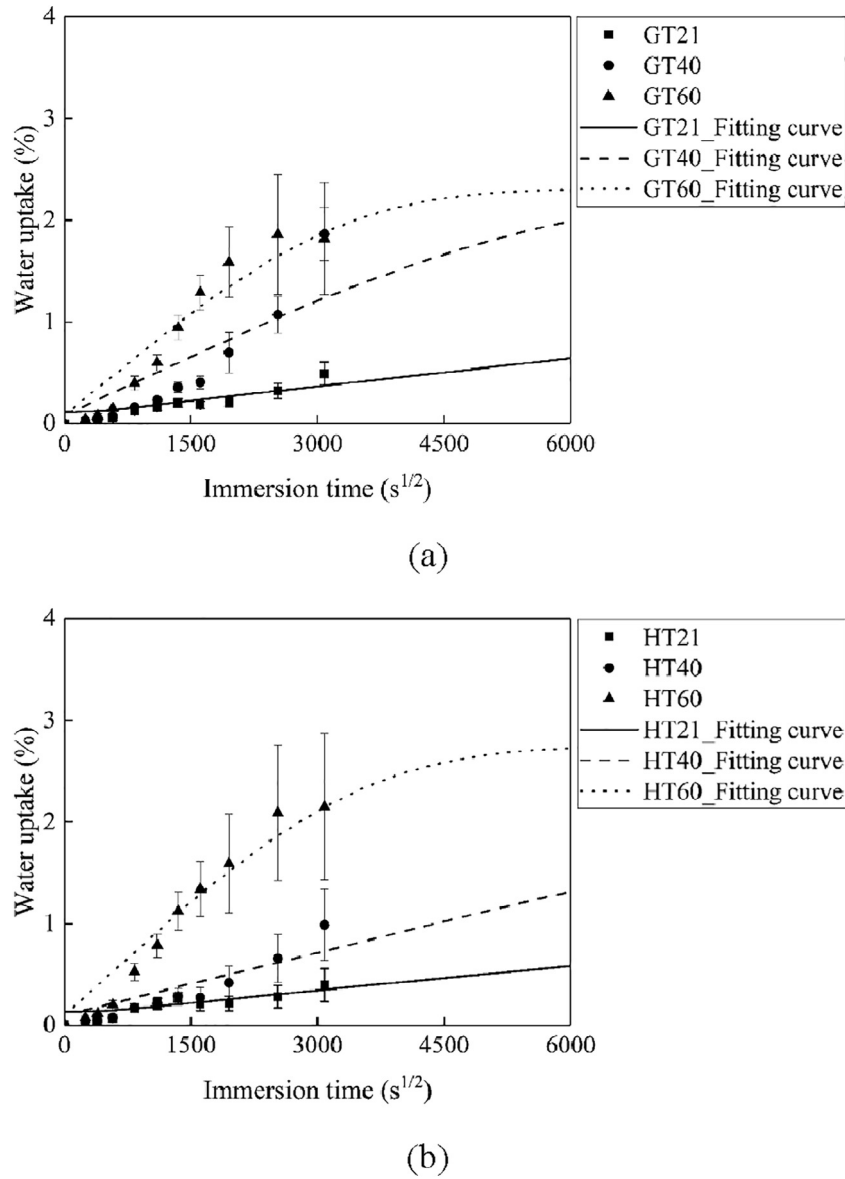


Fig. 5. Water uptake VS immersion time curves of GFRP (a) and HFRP (b) bars.

Table 4

Diffusivity coefficient and saturated water absorption of FRP bars.

Temperature (°C)	Diffusivity coefficient D_c ($10^{-13} \text{ m}^2/\text{s}$)		Saturated water absorption (%)	
	GFRP	HFRP	GFRP	HFRP
21 °C	0.08	0.04	2.31	2.75
40 °C	1.24	0.25	2.31	2.75
60 °C	3.85	3.38	2.31	2.75

to Eq. (5), the E_{ad} of GFRP and HFRP bars are determined to be 82.8 and 92.7, respectively. E_{ad} of HFRP bars is 12% larger than that of GFRP bars, indicating that the water molecular diffusion in GFRP bars should be much easier than that in HFRP bars.

The diffusivity coefficients and the water absorption of both GFRP and HFRP bars under two selected conditions are calculated by Eqs. (4) and (3), respectively, with the results shown in Table 5. The 1st selected condition is aged at 4.0 °C for 11 years and the 2nd selected condition is aged at 13.9 °C for 15 years. The two selected conditions

are the service conditions of GFRP bars in the Val-Alain Bridge and the Sierrita de la Cruz Creek Bridge, respectively [1,31].

3.2. Failure modes of GFRP and HFRP bars subject to short beam shear test

Fig. 7 shows the images of failed GFRP and HFRP bars subject to short beam shear test. Fig. 7a shows the images of control sample of GFRP and HFRP bars, within which crush and cracks can be seen. Under loading, the loading areas (central part) of control GFRP and

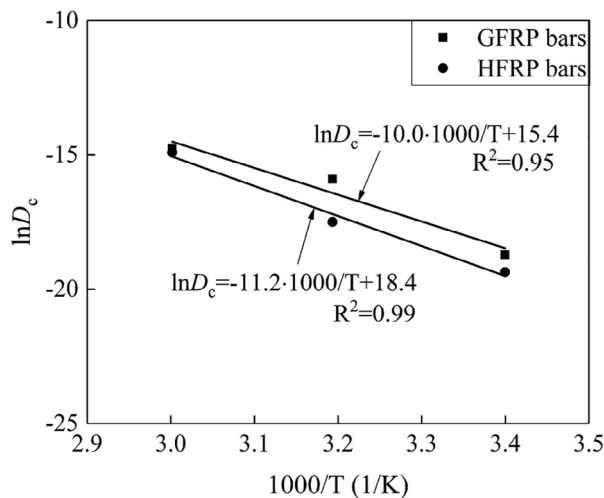


Fig. 6. Plots and fitting curves of diffusivity coefficient VS temperature.

HFRP bars were crushed first, and then cracks were formed along the neutral axis of bar. Both GFRP and HFRP bars finally failed owing to the primary crack along the neutral axis. Comparing Fig. 7a, b&e, it can be seen that the failure modes of both FRP bars immersed in alkaline solution at 21 °C, regardless of immersion time, are similar to that of control sample. The failure of FRP bars conditioned at 21 °C is primarily controlled by cracking at the neutral axis. This indicates that 21 °C conditioning does not significantly deteriorate the shear strength of both FRP bars.

As shown in Fig. 7c,d,f&g, smeared cracks were found in the outer surface of GFRP bars conditioned at 40 °C and 60 °C. And, the higher the conditioning temperature, the more the smeared cracks. With fast diffusion of water molecules and OH⁻ into GFRP bars at elevated temperature, deterioration of matrix and the interface between glass fiber and matrix occurs, leading to formation of smeared cracks in the outer surface of GFRP bars during the short beam shear test. However, even being conditioned at 40 °C and 60 °C, the failure of HFRP bars is mainly caused by the primary cracks along neutral axis, and the higher the temperature, the larger the open crack width. There was almost no smeared cracks found in the outer surface of HFRP bars.

3.3. Short beam shear test results

The calculated interlaminar shear strength ($\tau_{\max}$) of GFRP and HFRP bars conditioned at various temperature and for various periods are summarized in Table 6. Fig. 8 shows the plots of interlaminar shear strength ($\tau_{\max}$) with conditioning time.

As can be seen from Fig. 8 which shows the plots of interlaminar shear strength ($\tau_{\max}$) of FRP bars against conditioning time, the shear strengths of both GFRP and HFRP bars conditioned at 21 °C increases slightly until 70 days owing to the post-curing of matrix materials, and then decreases with increasing conditioning time. Compared to control samples, the shear strength of both GFRP and HFRP bars is reduced by 2% after 140 days immersion in alkaline solution. The slight degradation in shear strength of both GFRP and HFRP bars after 140 days

conditioning is resulting from slight deterioration of fiber, interface and matrix due to diffusion of water molecules and OH⁻. With conditioning temperature increasing, the diffusion of water molecules and OH⁻ and hence the deterioration of fiber, interface and matrix of FRP bars become faster, leading to faster degradation in the shear strength of both GFRP bars. When the conditioning temperature is 40 °C, the shear strengths of both FRP bars increase slightly until 30 days, and then start to decrease with increasing conditioning time at a larger rate. For the case of 40 °C, the shear strengths of both FRP bars decrease, without the increase stage, decreases with increasing conditioning time at the largest rate. Compared to the control samples, the shear strengths of GFRP bars conditioned at 40 °C and 60 °C for 140 days were reduced by 71% and 57%, respectively, while the reduction for HFRP bars are 12% and 34%, respectively.

As elevated temperature increases the diffusivity coefficient and hence accelerates the water molecule and OH⁻ diffusion and therefore deterioration, at a specified conditioning period, the shear strength of FRP bars, regardless of the type, decreases with increasing conditioning temperature. This is consistent with the failure modes of FRP bars conditioned at different temperatures as discussed in the above section. Compared with control GFRP bar sample, the control HFRP bars show similar shear strength. However, after being aged in alkaline solution, at a specified conditioning temperature, HFRP bars show a reduced strength reduction rate compared with GFRP bars. This could be attributed to the lower diffusivity coefficient of HFRP bars. Moreover, for a same conditioning case, the shear strengths of HFRP bars are higher than those of GFRP bars, especially for those long term and high temperature conditioning. For example, compared with GFRP, the shear strengths of HFRP conditioned at 40 °C and 60 °C for 100 days are 34% and 15% higher, respectively, indicating that addition of CFRP layer improve the durability of GFRP in alkaline environment. This difference in shear strength between GFRP and HFRP can be attributed to the differences in alkali resistance between carbon and glass fibers, and the differences in diffusivity coefficient between GFRP and HFRP bars. The degradation of both GFRP and HFRP bars in the shear strength is related to the water absorption in FRPs.

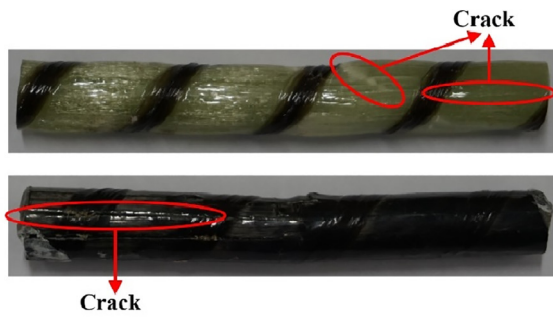
According to Eq. (3), the water absorption of both GFRP and HFRP bars versus conditioning time at various conditioning temperatures were calculated. The interlaminar shear strength of FRPs versus water absorption was plotted in Fig. 9, and fitting lines were made for results of GFRP and HFRP, respectively. It can be found that the shear strength of both GFRP and HFRP bars linearly decreases with increasing water absorption. And, the figure R^2 for the fitting lines of GFRP and HFRP bars are 0.69 and 0.83, respectively, indicating that the fitting lines could accurately predict the evolution of shear strength of FRP bars against the water absorption. This further proves that the shear strength degradation of both FRP bars immersed in alkaline solution are caused by water uptake and OH⁻ diffusion. Fig. 9 also shows that, against water absorption, the shear strength of GFRP bars reduces at a higher rate than that of HFRP bars. This indicates that, compared with GFRP bars, HFRP bars has better resistance to alkaline solution.

3.4. Degradation mechanism

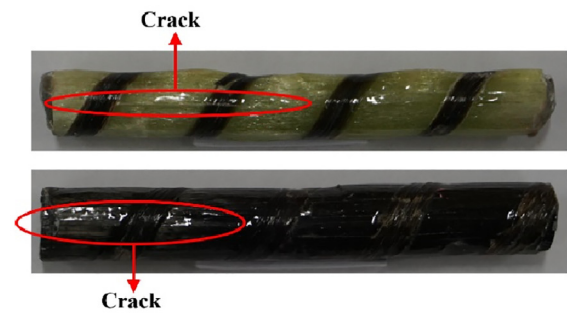
Fig. 10 shows the cross-section morphologies of FRP bars conditioned in alkaline solution for 140 days. Fig. 10a illustrates that the

Table 5
Diffusivity coefficient and water absorption of FRP bars at 4.0 °C and 13.9 °C.

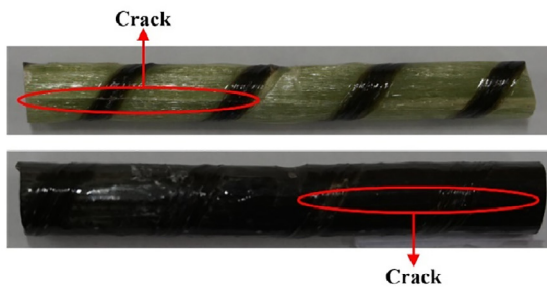
Conditions	Water absorption (%)		Diffusivity coefficient ($\times 10^{-16}$ m ² /s)	
	GFRP bars	HFRP bars	GFRP bars	HFRP bars
4.0 °C with 11 years	0.78	0.54	12.0	3.3
13.9 °C with 15 years	1.40	0.97	41.3	13.3



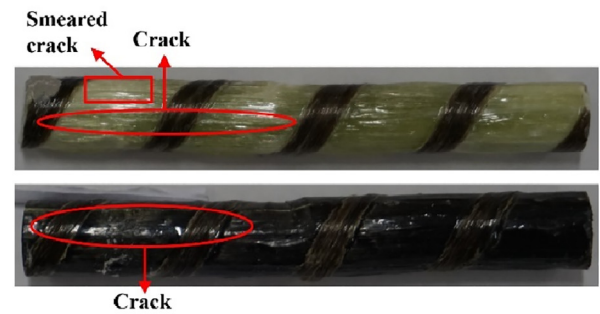
(a)



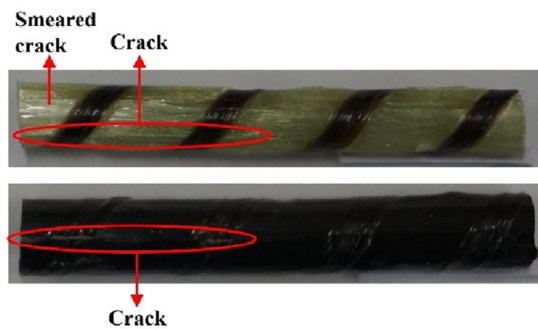
(f)



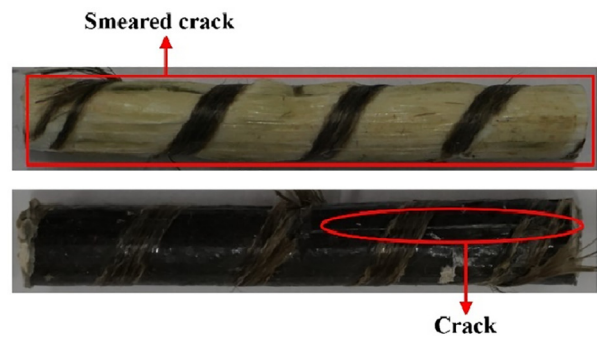
(b)



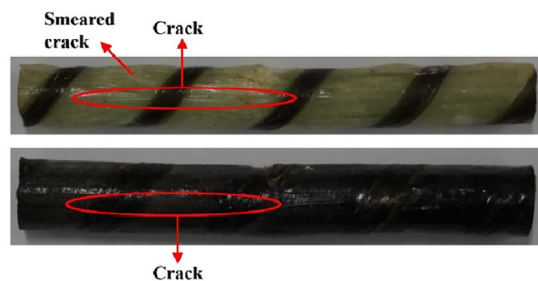
(g)



(c)



(f)



(d)

Fig. 7. Images of failed GFRP and HFRP bars immersed in alkaline solution, (a) control specimen, (b) 21 °C at 30 days, (c) 40 °C at 30 days, (d) 60 °C at 30 days, (e) 21 °C at 100 days, (f) 40 °C at 100 days, (g) 60 °C at 100 days.

interface region between glass fiber and matrix is intact for GFRP bars conditioned at 21 °C. For GFRP bars conditioned at 40 °C and 60 °C, the deterioration of the fiber–matrix interface region increases with

Fig. 7 (continued)

temperature increasing, as shown in Fig. 10b&c. Loss of matrix materials and pores were also found at the outer layer of GFRP bars owing to the hydrolysis of epoxy resin and dissolution of small molecular monomer. The deterioration of GFRP bars is found from the outer surface to center.

Fig. 10d shows relative intact interface region between the internal GFRP bar and the external CFRP coat of HFRP. A few deterioration are found occurring within GFRP close to the GFRP/CFRP interface, while the CFRP region is still intact. This can be attributed to the better resistance of carbon fiber to water and OH⁻. As shown in Fig. 10e, the deterioration of GFRP close to the GFRP/CFRP interface increases after being conditioned at 60 °C, and a lot of pores are even found within GFRP close to the CFRP/GFRP interface.

The morphologies of the fracture surface of both GFRP and HFRP bars subject to short beam shear test were observed with SEM. Fig. 11 shows the SEM image of fracture surface of GFRP and HFRP bars immersed in alkaline solution for 0 days and 140 days. Large

Table 6
Shear strengths of GFRP and HFRP bars aged in alkaline solution.

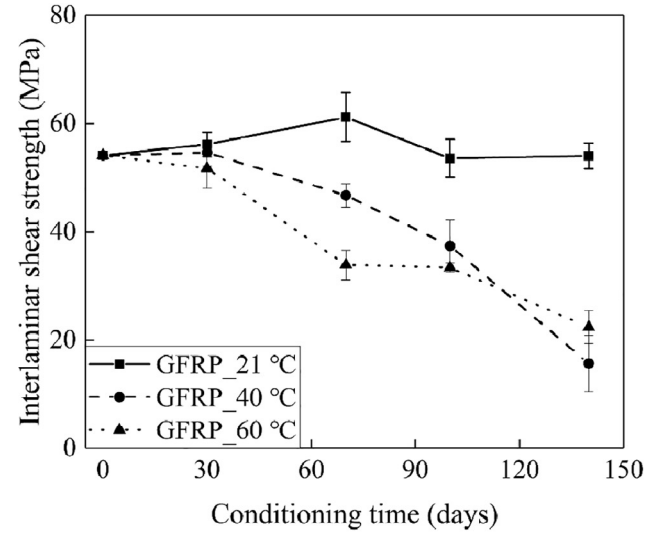
Specimens	$\tau_{\max}^a$ (MPa)	CV ^b
GD0	54.1	0.7
GT21D30	54.9	0.5
GT21D70	61.2	4.5
GT21D100	53.6	3.5
GT21D140	52.3	2.0
GT40D30	54.7	2.4
GT40D70	46.7	2.2
GT40D100	37.4	4.8
GT40D140	15.7	5.2
GT60D30	51.8	3.8
GT60D70	33.9	2.7
GT60D100	28.9	7.8
GT60D140	23.1	2.9
HD0	55.3	1.4
HT21D30	57.5	1.6
HT21D70	62.2	1.5
HT21D100	55.3	5.0
HT21D140	57.3	5.5
HT40D30	59.4	1.8
HT40D70	53.3	5.2
HT40D100	52.4	5.9
HT40D140	51.1	4.0
HT60D30	53.5	3.3
HT60D70	48.1	2.5
HT60D100	42.6	2.5
HT60D140	37.8	10.1

Note: a- $\tau_{\max}$ is the interlaminar shear strength.

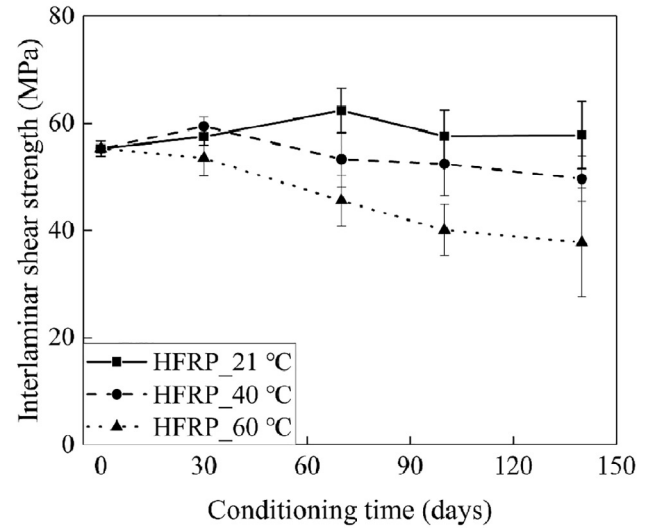
b- CV is coefficient of variation.

amount of residual matrix materials are found adhering on the fiber surface of control GFRP samples, as shown in Fig. 11a. Fig. 11b,c&d show that the amount of residual matrix materials on fiber surface of GFRP decreases with conditioning temperature increasing. And, there is even no residual matrix materials found on glass fiber of GFRP bars conditioned at 60 °C. This indicates that the deterioration of fiber–matrix interface becomes more and more severe with increasing conditioning temperature.

Fig. 11e-h show the fracture surface morphologies of HFRP bars. Similar to GFRP bars, large amount of matrix materials are also found on both glass fibers and carbon fibers within HFRP bars. The amount of residual matrix materials on glass fiber decreases with increasing conditioning temperature, while a lot of residual matrix materials can still be found on carbon fiber within the HFRP bars conditioned at 60 °C in Fig. 11h. This indicates that the carbon fiber–matrix interface shows a better alkaline resistance than the glass fiber–matrix



(a)



(b)

Fig. 8. Shear strengths versus conditioning time for GFRP (a) and HFRP (b) bars conditioned at different temperatures.

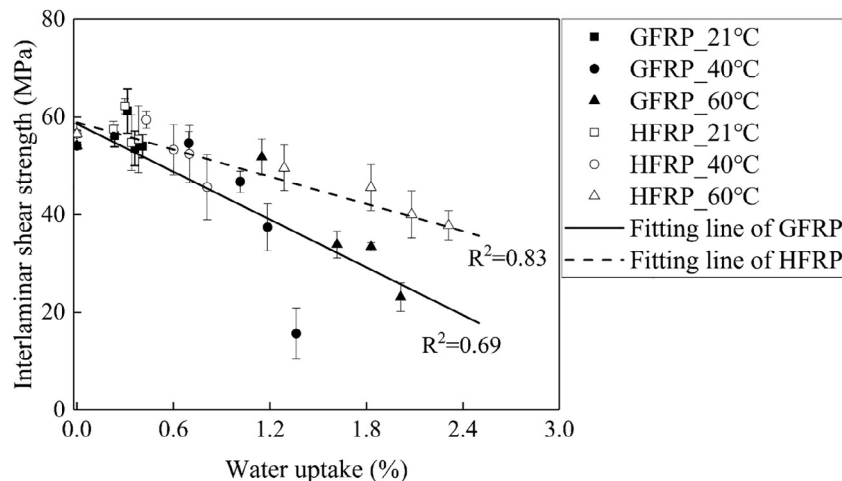


Fig. 9. Shear strength of GFRP and HFRP bars versus water uptake.

interface. The degradation in shear strength of HFRP bars is mainly caused by the deterioration of the glass-fiber matrix interface close to the CFRP/GFRP interface. The OM and SEM observation results are consistent to the short beam shear test results of both GFRP and HFRP bars.

3.5. Prediction of long-term behavior of FRP bars

The long-term durability of GFRP and HFRP bars can be predicted by the accelerated aging of FRP bars with Arrhenius degradation theory. The assumption of the Arrhenius degradation theory is that the rate of degradation of FRPs is accelerated with environmental temperature increasing, and the single dominant degradation mechanism of FRPs does not change at the exposure duration [32]. According to the previous discussion, the deterioration of both GFRP and HFRP bars

was dominated by hydrolysis of epoxy resin and the debonding between glass fiber and matrix in alkaline solution, and the degradation of interlaminar shear strength increased with environmental temperature. The Arrhenius degradation theory was applied, and the degradation rate can be expressed as the below Eq. (6):

$$k = B \exp(-E_{am}/R_1 T) \quad (6)$$

where k is the degradation rate (1/time), B is the constant of the material and degradation process, E_{am} is the activation energy of degradation in FRPs (J/mol). T is Kelvin temperature (K).

The logarithm of $1/k$ can be expressed as a linear function of $1/T$ with the slope of E_{am}/R_1 , and the Eq. (6) is transformed to be:

$$\ln\left(\frac{1}{k}\right) = \frac{E_{am}}{R_1} \frac{1}{T} - \ln B \quad (7)$$

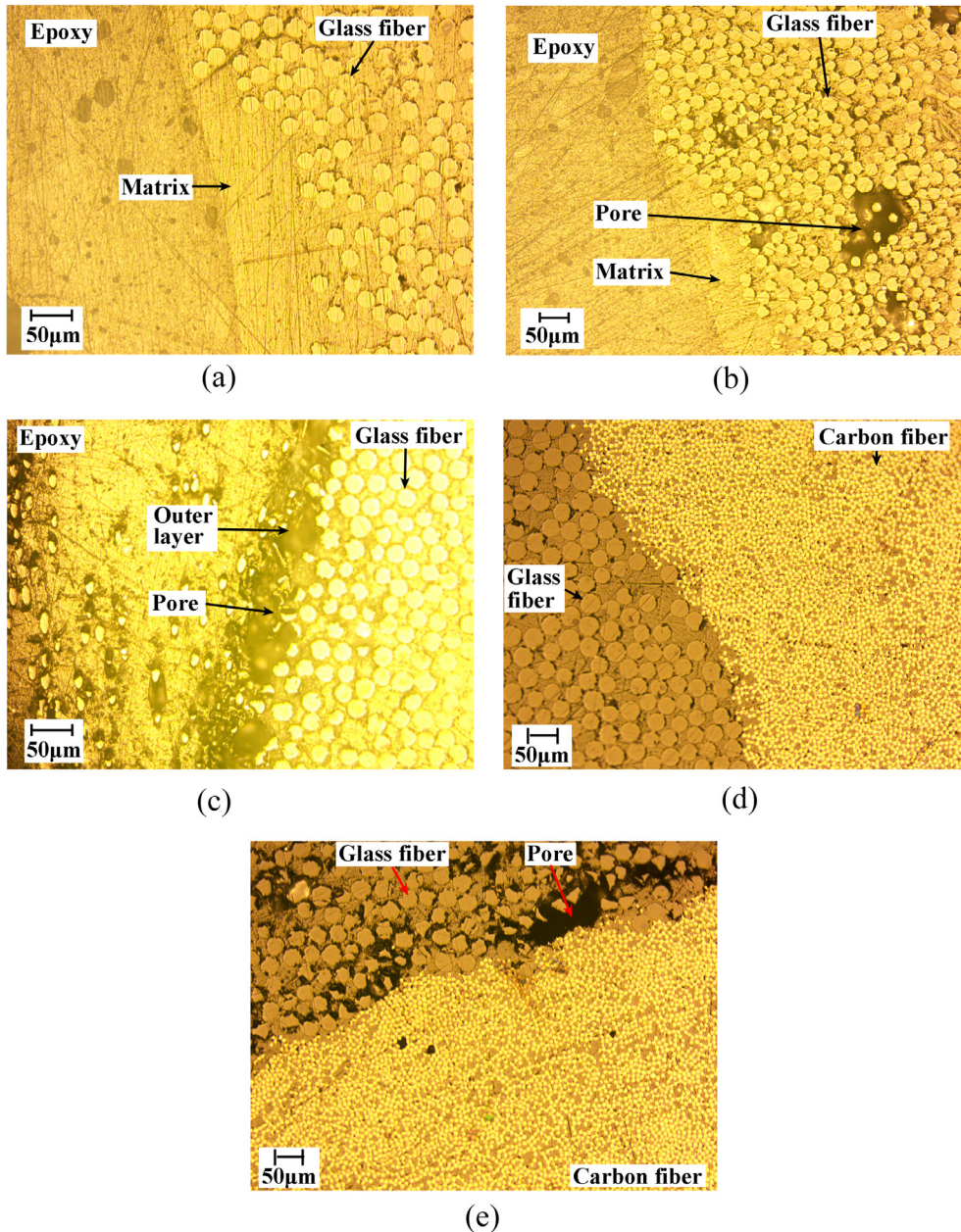
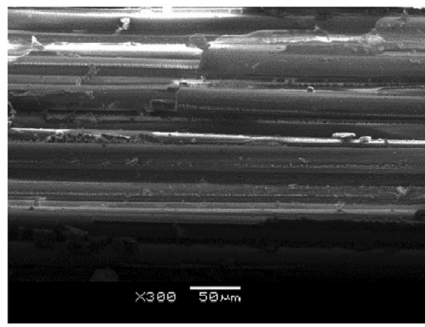
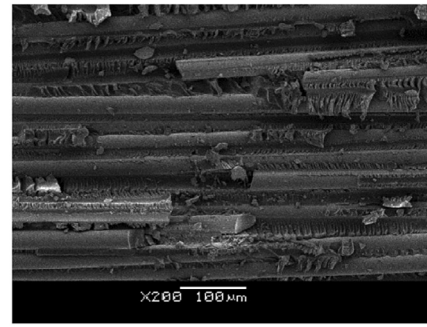


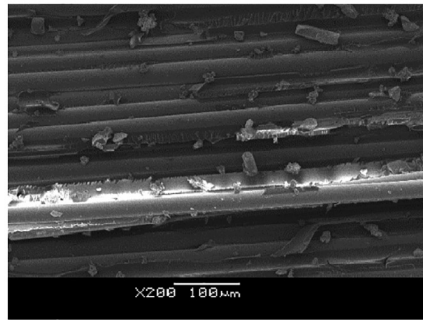
Fig. 10. OM images of the cross-section of GFRP and HFRP bars aged for 140 days, (a) GFRP at 21 °C, (b) GFRP at 40 °C, (c) GFRP at 60 °C, (d) HFRP at 40 °C and (e) HFRP at 60 °C.



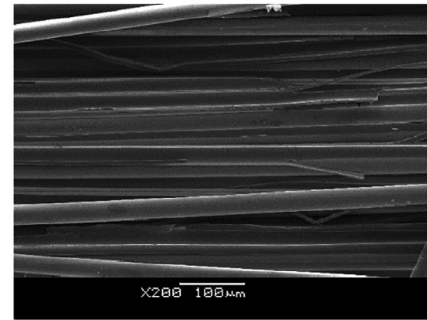
(a)



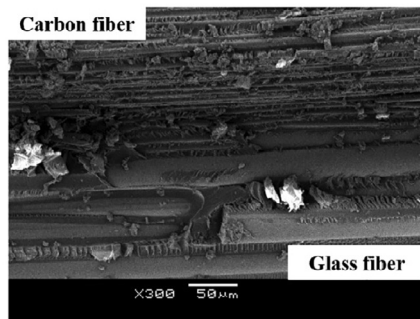
(b)



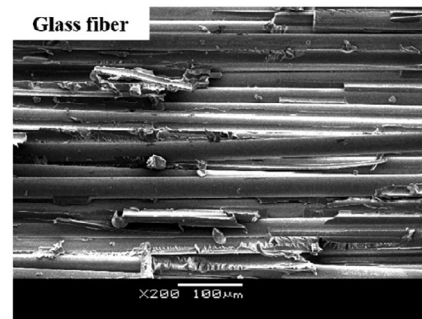
(c)



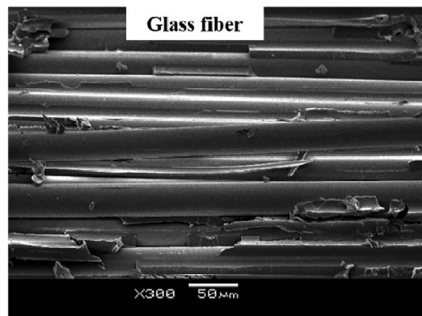
(d)



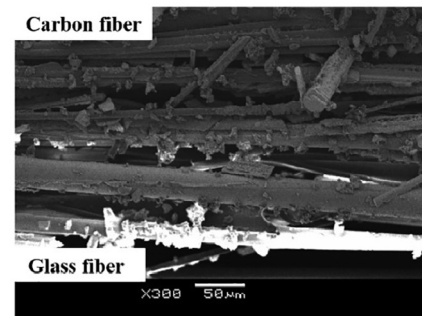
(e)



(f)



(g)



(h)

Fig. 11. SEM image of the fracture surface of GFRP and HFRP bars aged for 0 and 140 days, (a) Control GFRP, (b) GFRP at 21 °C, (c) GFRP at 40 °C, (d) GFRP at 60 °C, (e) Control HFRP, (f) HFRP at 40 °C, (g) HFRP at 60 °C and (h) HFRP at 60 °C.

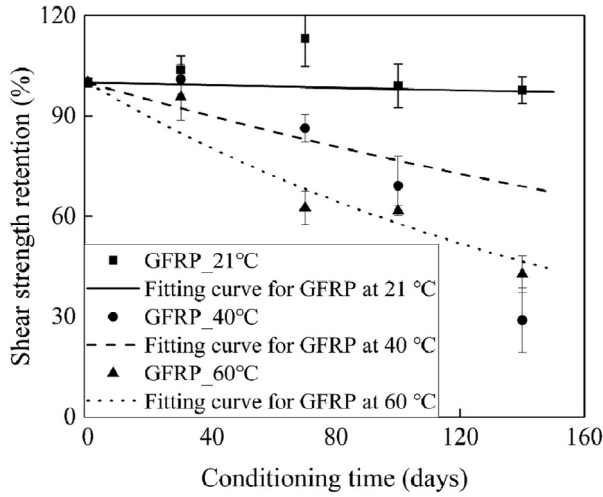
Previous researches proved that the below Eq. (8) can be used to define the relationship between shear strength retention of FRP bars and conditioning time [6,8].

$$Y = 100\exp(-t/\tau) \quad (8)$$

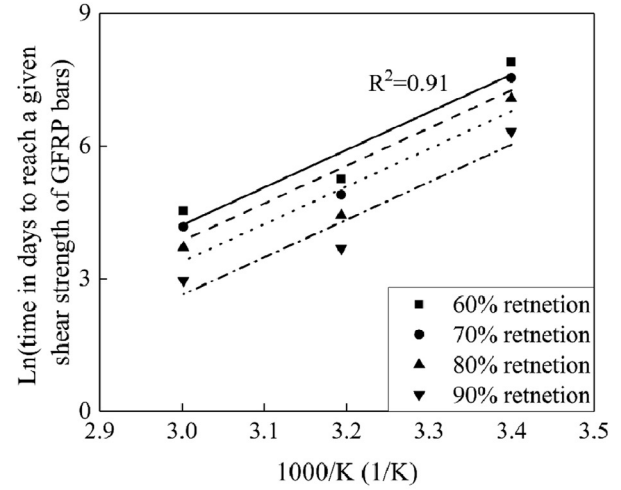
where Y is the shear strength retention (%), t is the exposure time, τ is the fitted parameter. Fig. 12 compares the fitting curves and the experimental results of GFRP and HFRP bars. The parameters fitted by Eq. (8) were listed in Table 7. τ of both GFRP and HFRP bars decreases with an

increase of environmental temperature. It is worth noting that the effects of post-curing at 21 °C on the interlaminar shear strength was ignored. It was assumed that the alkaline solution at 21 °C reduced the interlaminar shear strength with conditioning time. In the case of specimens at 21 °C, the tested results at 40 and 70 days were removed when Eq. (8) was applied.

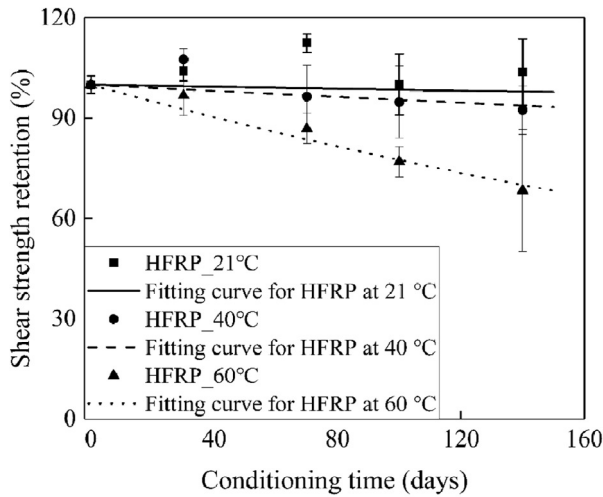
The 60%, 70%, 80% and 90% of the shear strength retention at 21 °C, 40 °C and 60 °C are determined by Eq. (8) with τ values shown in Table 7. The results of $\ln(t)$ vs. $1000/T$ are plotted in Fig. 13. Eq. (7) is



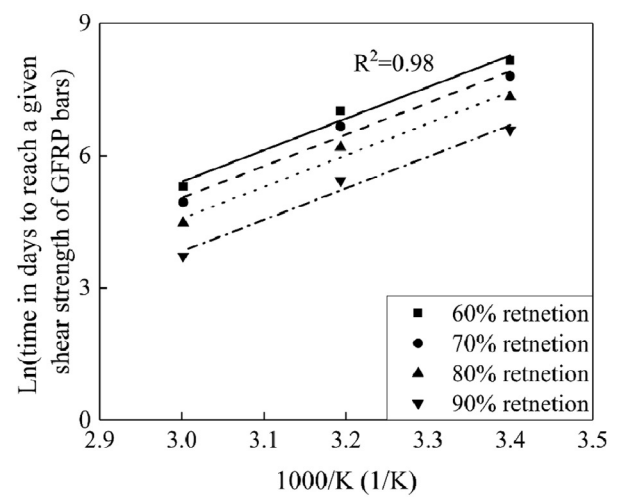
(a)



(a)



(b)



(b)

Fig. 12. Fitting curves and experimental results of GFRP (a) and HFRP (b) bars.

Table 7
Parameters fitted by Eq. (8).

Type	Temperature (°C)	τ
GFRP	21	5292
	40	377
	60	183
	21	6860
HFRP	40	2185
	60	314

Fig. 13. Arrhenius plots of shear strength retention of GFRP bars (a) and HFRP bars (b) aged in alkaline solution.

Table 8
Parameters fitted by Eq. (7).

Type	Shear strength retention (%)	E_{am}/R_1	R^2
GFRP	60	8509	0.91
	70	8509	0.91
	80	8509	0.91
	90	8509	0.91
HFRP	60	7165	0.98
	70	7165	0.98
	80	7165	0.98
	90	7165	0.98

Table 9
Time-shift factor for FRPs in alkaline solution.

Type	Temperature (°C)	Time-shift factor (TSF)			
		Hangzhou, China (17.0 °C)	Sierrita de la Cruz Creek Bridge (13.9 °C)	Waterloo Creek Bridge (9.9 °C)	Val-Alain Bridge (4.0 °C)
GFRP	21	1.5	2	3.1	5.9
	40	8.6	11.8	18	34.1
	60	44	60.5	91.9	174.3
HFRP	21	1.4	1.8	2.6	4.5
	40	6.1	8	11.4	19.5
	60	24.2	31.6	45	77.1

adopted to fit the dots, and the straight lines are exactly parallel to each other in Fig. 13. The parameters fitted by Eq. (7) are listed in Table 8.

Based on Eq. (6), the time-shift factor (TSF) for the apparent horizontal shear strength to reach the same value (represented by c) at temperatures T_1 and T_0 can be obtained from the above Arrhenius plots Fig. 14. The TSF can be expressed as [33]:

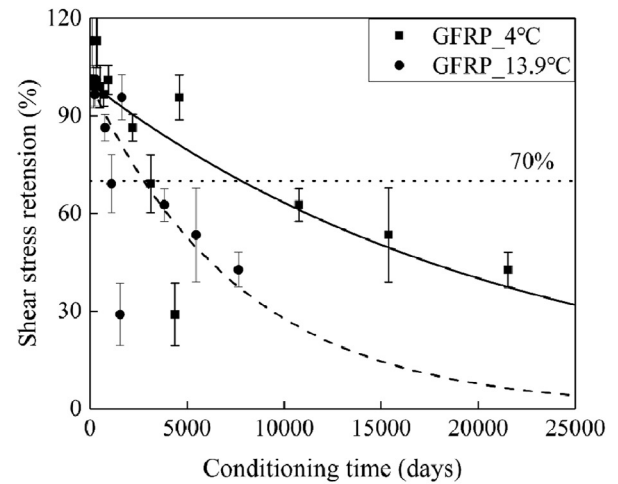
$$\text{TSF} = \frac{t_0}{t_1} = \frac{c/k_0}{c/k_1} = \frac{k_1}{k_0} = \frac{\text{Bexp}(-E_{\text{am}}/R_1 T_1)}{\text{Bexp}(-E_{\text{am}}/R_1 T_0)} = \exp\left[\frac{E_{\text{am}}}{R_1} \left(\frac{1}{T_0} - \frac{1}{T_1}\right)\right] \quad (9)$$

In the present study, four typical environmental temperatures ranging from 4.0 °C to 17 °C are selected to predict the long-term performance of GFRP and HFRP bars in alkaline environment. The four selected annual mean temperatures include 17.0 °C in Hangzhou, 13.9 °C for Sierrita de la Cruz Creek Bridge in America [31], 9.9 °C for Waterloo Creek Bridge in Canada [34] and 4.0 °C for Val-Alain Bridge in Canada [1]. The Sierrita de la Cruz Creek Bridge, Val-Alain Bridge and Waterloo Creek Bridge are reinforced with GFRP bars and are still in service. According to Eq. (9), the TSF with reference temperature T_0 equal to the four selected annual mean temperatures are calculated and listed in Table 9.

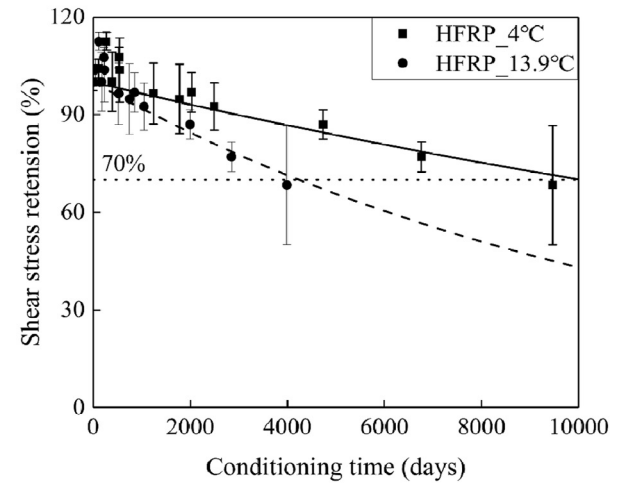
Once the TSF values for 21 °C, 40 °C and 60 °C are obtained, the master curves for shear strength retention versus conditioning time at four selected temperatures could be obtained by multiplying the conditioning times at 21 °C, 40 °C and 60 °C with corresponding TSF values. Fig. 14 shows the prediction curves for conditioning GFRP and HFRP bars at 4 °C in the Val-Alain Bridge region and 13.9 °C in the Sierrita de la Cruz Creek Bridge region. All the fitted results for the master curves of FRP bars conditioned in alkaline solution are listed in Table 10.

ACI 440.1R-15 recommends that the environmental reduction factor (C_E) is adopted to consider the mechanical properties degradation of FRPs [35]. C_E is set to be 70% for GFRP bars in concrete exposed to earth and weather. The straight line of shear strength retention equal to 70% was plotted in Fig. 14. The exposure time to reach 70% of the shear strength retention is determined by the intersection of straight line of shear strength retention equal to 70% and the predicted curve in Fig. 14, and the results are listed in Table 11. It takes 5.6 to 21.4 years and 9.1 to 27.4 years for GFRP and HFRP bars, respectively, to make the shear strength retention reach 70% in alkaline solution at the temperature from 4.0 °C to 17 °C. This indicates that the CFRP coat on GFRP bars can improve the durability of GFRP bars.

Table 12 compares the predicted shear strength results and the measured shear strength results (field results) for the GFRP bars in the Sierrita de la Cruz Creek Bridge and the Val-Alain Bridge. It is reported that the shear strength of GFRP bars in the Val-Alain Bridge (4 °C) is reduced by 14% [1] after service for 11 years, while the shear strength of GFRP bars in the Sierrita de la Cruz Creek Bridge (13.9 °C) is increased by 4% after service for 15 years [31]. However, according to the prediction, the shear strengths of GFRP bars are decreased by 15% and 47% after 11 years at 4.0 °C and 15 years at 13.9 °C, respectively. Compared with the field results, the prediction results overestimate the degradation of the shear strength of GFRP bars. The slight



(a)



(b)

Fig. 14. Prediction curves for (a) GFRP bars at 13.9 °C and 4.0 °C and (b) HFRP bars at 13.9 °C and 4.0 °C.

increase in shear strength of GFRP bars in Sierrita de la Cruz Creek Bridge may be caused by the post curing of matrix materials at a relative high annual mean temperature (e.g., 13.9 °C).

Table 13 lists the measured shear strength (field results) of GFRP bars in the Val-Alain Bridge and the predicted shear strength results of GFRP and HFRP bars against water absorption. The relationship between water absorption and conditioning time is established with Eq. (3) and diffusion parameters in Table 5. The shear strength retention of both GFRP and HFRP bars against water absorption is deter-

Table 10

Coefficients for the regression equations for the master curves of FRP bars aged in alkaline solution.

Type	City	Annual mean temperature (°C)	τ	R^2
GFRP	Hangzhou in China	17 °C	5716	0.64
	Sierrita de la Cruz Creek Bridge in USA	13.9 °C	7782	0.64
	Waterloo Creek Bridge in Canada	9.9 °C	11,720	0.64
	Val-Alain Bridge in Canada	4.0 °C	21,877	0.64
HFRP	Hangzhou in China	17 °C	9162	0.77
	Sierrita de la Cruz Creek Bridge in USA	13.9 °C	11,870	0.77
	Waterloo Creek Bridge in Canada	9.9 °C	16,692	0.77
	Val-Alain Bridge in Canada	4 °C	28,134	0.77

Table 11

Long-term predication results of FRP bars at different annual mean temperatures.

City	Annual mean temperature (°C)	Time in years to reach 70% shear strength retention	
		GFRP	HFRP
Hangzhou in China	17 °C	5.6	9.1
Sierrita de la Cruz Creek Bridge in USA	13.9 °C	7.7	11.5
Waterloo Creek Bridge in Canada	9.9 °C	11.4	16.4
Val-Alain Bridge in Canada	4 °C	21.4	27.4

mined according to Fig. 9. According to above discussion, the degradation in shear strength of GFRP bars is related to water absorption. In the Val-Alain Bridge, the relative humidity of concrete is approximate 80%, and the pH of concrete pore solution is in a range of 12 to 13. The 11 years water absorption of GFRP bars in Val-Alain Bridge is measured to be 0.14% [1]. In contrast, for GFRP and HFRP bars immersed in alkaline solution, 33 days and 5 days are needed respectively to reach a water absorption of 0.14%. This indicates that the higher the relative humidity, the faster the diffusion of water molecules. Compared to the 86% shear strength retention of GFRP bars in the Val-Alain Bridge, the shear strength retentions of GFRP and HFRP bars aged in alkaline solution are larger. The reason may be that the long-term hydroxylation of matrix in field is much more significant than the short-term hydroxylation of matrix in alkaline solution. The water absorption of GFRP and HFRP bars immersed in alkaline solution for 11 years achieve 0.78% and 0.54%, respectively, which are much higher than the water absorption of GFRP bars in the Val-

Alain Bridge. And, the water absorption of 0.78% and 0.54% induce 22% and 9% reduction in shear strength of GFRP bars and HFRP bars, respectively, according to Fig. 9. For a specified service/aging period, the degradation of GFRP in alkaline solution is more severe than that of GFRP bars in the Val-Alain Bridge. This is because the water uptake of GFRP bars in alkaline solution is much larger than that of GFRP bars in concrete.

In comparisons of the results shown in the present study and the filed results, the prediction results obtained shown in Table 12&13 according to the present study overestimate the degradation of mechanical properties of both GFRP and HFRP bars in field. It is clear that the long-term prediction results are too conservative when compared with the field results. As the mechanical properties degradation of both GFRP and HFRP is mainly caused by the water absorption which depends on the relative humidity in the service environment of FRPs [36,37], the humidity and temperature of concrete in service condition should hence be taken into account in the further study.

Table 12

Comparison between the prediction results and the field results of FRP bars.

City	Annual mean temperature (°C)	Service time (years)	Shear strength retention (%)			
			Field results		Predicted results	
			Value	CV	GFRP	HFRP
Val-Alain Bridge in Canada	4.0 °C	11	86	0.02	85	88
Sierrita de la Cruz Creek Bridge in USA	13.9 °C	15	104%	–	53	66

Table 13

Comparison between long-term prediction results with water absorption and field results of FRP bars.

Water absorption (%)	Time to reach given water absorption		Shear strength retention (%)		
	GFRP in alkaline solution	HFRP in alkaline solution	GFRP bars in Val-Alain Bridge after 11 years	GFRP bars in alkaline solution	HFRP bars in alkaline solution
0.14	33 days	5 days	86%	96%	98%
0.54	–	11 years	–	–	91%
0.78	11 years	–	–	78%	–

4. Conclusion

The present study investigated the durability, in terms of interlaminar shear strength, of GFRP and HFRP bars aged in alkaline solution at different temperatures. Based on the results, the following can be highlighted:

- (1) The water absorption behaviors of both GFRP and HFRP bars follow the Fickian law. The diffusivity coefficient of water molecule in HFRP bars is smaller than that in GFRP bars, but the saturated water absorption of HFRP bars is larger than that of GFRP bars.
- (2) Both GFRP and HFRP bars subject to short beam shear test failed due to the primary crack along the neutral axis of bars. Under same aging condition, the shear strength retention of HFRP bars is larger than that of GFRP bars. The CFRP coat delays the mechanical properties degradation of GFRP bars.
- (3) Water and OH⁻ penetration deteriorates the fiber-matrix interface region in GFRP bars. The deterioration of GFRP bars is from the outer surface to center. Deterioration of HFRP bars mainly occurs within the internal GFRP close to the CFRP/GFRP interface.
- (4) The long-term shear strength reduction of GFRP and HFRP bars conditioned in alkaline solution were predicted using the Arrhenius theory, and the prediction results significantly overestimate the shear strength reduction of GFRP bars service in real structures.
- (5) A more realistic model for predicting the degradation of FRP bars should be developed by considering the real temperature and humidity ranges.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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